Chi-Sheng (Michael) Chen (陳麒升)

Machine Learning for Physiological Time Series | Clinical Biosignals & Foundation Models | EEG/BCI | Quantum ML

+1 (351) 244-1084 | m50816m50816@gmail.com | LinkedIn | GitHub | Publication
Nationality: Taiwan | Based in Cambridge, MA, USA

Chi-Sheng (Michael) Chen builds machine learning for physiological time series along one arc — from representation, to foundation models, to clinical deployment. He develops interpretable, clinician-interrogable frequency-domain and geometric representations of clinical biosignals (FreqLens, SPD Token Transformer) and scales them into biosignal foundation models (Large Cognition Model). His EEG-based depression-treatment model is in routine outpatient use at Taipei Veterans General Hospital's Precision Depression Intervention Center (PreDIC), and at Harvard/BIDMC he builds multimodal EMS trauma-triage pipelines. He has 48 publications, 450+ citations (h-index 13, i10-index 16), ICML 2026 Gold Reviewer recognition, and reviewer service across 35 journals (incl. TPAMI, NSR, npj Quantum Information). As a secondary methodological line, he explores hybrid quantum-classical architectures for sequence modeling (QASA, QEEGNet).

Research & Professional Experience

Harvard Medical School & Beth Israel Deaconess Medical Center — MA, USA
Research Affiliate, Advisor: Prof. Dr. Gabriel Brat — *Nov 2024 – Present*

- Developing real-time EMS triage pipeline using multimodal AI for trauma prediction.
- Collaborating with surgeons on AI-assisted decision support systems.

OpenAI (contractor via Mercor) — Remote (US)
AI Trainer — *Mar 2025 – Oct 2025*

- Performed high-complexity AI data labeling and evaluation tasks, including instruction following, multimodal reasoning, and safety alignment.
- Contributed to Reinforcement Learning from Human Feedback (RLHF) pipelines by providing high-quality comparative judgments, preference rankings, and model-behavior assessments.

Omnis Labs — Remote
Co-Founder — *Sep 2024 – Present*

- AI-driven DeFi liquidity positioning protocol with automated market maker (AMM) strategies and backtesting systems.
- Achieved 50%–120% base fee APR on WBTC/USDC and ETH/USDC pairs (Katana on SushiSwap) using time-series model-driven AMM strategies.

Neuro Industry, Inc. — CA, USA
Researcher, Co-Founder & CTO — *Mar 2024 – Jan 2025*

- Built GCP-based MLOps pipeline, enabling training on 10K+ EEG samples.
- 2 research papers on quantum machine learning on EEG signal processing.
- 1 research paper on EEG-to-image using diffusion model.

Department of Computer Science, National Yang Ming Chiao Tung University — Hsinchu, Taiwan
Research Assistant, Advisor: Prof. Chun-Shu Wei — *Dec 2023 – Aug 2024*

- Developed MUSE, a multimodal contrastive learning framework for EEG-based image recognition (31 citations).
- Research on nonstationary EEG dynamics and foundation models for brain-computer interfaces.

MediaTek Inc. — Hsinchu, Taiwan
Digital IC R&D Engineer — *Dec 2021 – Oct 2023*

- Responsible for microprocessor IP development for flagship 5G smartphones' display and AMBA SoC implement.
- Hardware virtualization architecture RTL design and IP verification with UVM and SystemVerilog.

Department of Surgery, National Taiwan University Hospital — Taipei, Taiwan
Research Assistant, Advisor: Dr. Shuo-Lun Lai — *Jul 2021 – Sep 2021*

- Deep learning applications on surgery automation with YOLO-based model.
- Developed a prototype for a surgical smoke evacuation system.

Department of Psychiatry, Taipei Veterans General Hospital — Taipei, Taiwan
Research Assistant, Advisor: Prof. Dr. Cheng-Ta Li — *Sep 2019 – Jun 2021*

- Collaborated with researchers and clinicians to identify patterns and trends in brainwave activity related to major depressive disorder.
- Utilized machine learning algorithms to develop predictive models based on real-world brainwave data.

Max Planck Institute for Chemical Physics of Solids (MPI CPfS) — Dresden, Germany
Research Internship, Advisor: Dr. Alexander Komarek & Dr. Li Zhao — *Jul 2018 – Sep 2018*

- Searching new possible unconventional superconductors among Co-based quaternary chalcogenides with diamond like structure $CuInCo_2A_4$ / $AgInCo_2A_4$ (A = Te, Se, S).

Education

National Taiwan University (NTU) Taipei, Taiwan

Master of Science in Biomedical Electronics and Bioinformatics ***GPA: 3.74/4.30*** *Sep 2019 – Jun 2021*
Advisor: Prof. Dr. Cheng-Ta Li, MD & Prof. Chung-Ping Chen

- Machine learning application on non-stationary time series data.
- The research EEG model for TMS pattern personalized prediction is currently being applied to real outpatient patients in the Precision Depression Intervention Center (PreDIC), Taipei Veterans General Hospital.

National Chiao Tung University (NCTU) Hsinchu, Taiwan

Bachelor of Engineering in Electrophysics *Sep 2015 – Jun 2019*
Bachelor of Science in Interdisciplinary Science Degree Program

- Researching the oxide processing techniques.
- Researching the biomedical signal processing on traditional Chinese medicine data.

Skills

AI/ML: Python, PyTorch, HuggingFace, Scikit-learn, MNE, MLflow, R, MATLAB

Systems: C/C++, JavaScript, Rust, MySQL, Solidity, Git, Docker, GCP, AWS

Hardware: Verilog, SystemVerilog, UVM, HSPICE, Genus, Verdi, SG-Lint, Perl

Languages: Chinese (Native), English (Fluent)

Selected Publications (450+ cites, h-index 13, i10-index 16)

[Peer-Reviewed - First/Co-first Author]

[Journal]

Chen, C.-S., et al. "Exploring the potential of qeegnet for cross-task and cross-dataset electroencephalography encoding with quantum machine learning" Journal of Signal Processing Systems (2025)

Chen, C.-S., Kuo, E.-J. "Quantum Adaptive Self-Attention for Quantum Transformer Models" Quantum Science and Technology (2026), https://doi.org/10.1088/2058-9565/ae8b21 .

Chen, C.-S., et al. "Unraveling quantum environments: Transformer-assisted learning in Lindblad dynamics" Physical Review A (2025), https://doi.org/10.1103/gsxk-45mk .

Chen, C.-S., et al. "Exploring the Potential of Electroencephalography Signal–Based Image Generation Using Diffusion Models: Integrative Framework Combining Mixed Methods and Multimodal Analysis" JMIR MI (2025), https://doi.org/10.2196/72027 .

Chen, C.S., et al. "Improving fine-grained food classification using deep residual learning and selective state space models" PloS one (2025), https://doi.org/10.1371/journal.pone.0322695 .

[Conference]

Chen, C.S., et al. "Quantum Reinforcement Learning-Guided Diffusion Model for Image Synthesis via Hybrid Quantum-Classical Generative Model Architectures" 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), doi: 10.1109/ICASSP55912.2026.11461991 .

Chen, C.S., et al. "Quantum Adaptive Self-Attention for Financial Rebalancing: An Empirical Study on Automated Market Makers in Decentralized Finance" 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), doi: 10.1109/ICASSP55912.2026.11460972 .

Chen, C.S., et al. "Quantum Multimodal Contrastive Learning Framework" 2025 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), doi: 10.1109/ICASSP49660.2025.10889504, arXiv:2408.13919 .

Chen, C.S., et al. "QEEGNet: Quantum Machine Learning for Enhanced Electroencephalography Encoding" IEEE International Workshop on Signal Processing Systems (SiPS) 2024, doi: 10.1109/SiPS62058.2024.00035, arXiv:2407.19214 .

Chen, C.S., Liu, H.Y. "Predicting Poets' Origins from Verse: A Computational Analysis of Regional Linguistic Fingerprints in the Complete Tang Poems" Digital Humanities 2026 (DH2026), Alliance of Digital Humanities Organizations (ADHO), Short Paper.

[Workshop]

Chen, C.S., Brat, G.A. "Measuring Browser Webcam Gaze Honestly: A Capture-Clock Methodology and Open Reference Implementation" 4th MICCAI Workshop on Data Engineering in Medical Imaging (DEMI 2026), Poster, arXiv:2608.11566.

Chen, C.S., et al. "Q-DPTS: Quantum Differentially Private Time Series Forecasting via Variational Quantum Circuits"IEEE GLOBECOM Workshop 2025, arXiv:2508.05036.

Chen, C.S., et al. "Quantum Reinforcement Learning Trading Agent for Sector Rotation in the Taiwan Stock Market" IEEE QCE QCRL Workshop 2025, arXiv:2506.20930.

[Peer-Reviewed - Contributing Author]

Li, C.T., **Chen, C.S., et al. "Prediction of antidepressant responses to non-invasive brain stimulation using frontal electroencephalogram signals: Cross-dataset comparisons and validation"** Journal of Affective Disorders (2023), https://doi.org/10.1016/j.jad.2023.08.059 .

Lai, S.L., **Chen, C.S., et al. "Intraoperative Detection of Surgical Gauze Using Deep Convolutional Neural Network"** Ann Biomed Eng 51, 352–362 (2023), https://doi.org/10.1007/s10439-022-03033-9 .

Lai, S.L., Chou, Y.C., **Chen, C.S., et al. "Detection and tracking of a gauze sponge in minimally invasive surgery using a YOLO and R-CNN based model"** Medical & Biological Engineering & Computing (2025), https://doi.org/10.1007/s11517-025-03471-2 .

Liu, H.Y., **Chen, C.S.** "Gendered Language in Tang Poetry: A Sociolinguistic and Digital Humanities Analysis Based on the Quan Tang Shi." 16th International Conference of Digital Archives and Digital Humanities (DADH 2025).

Takegami D., et al.**"Direct imaging of valence orbitals using hard x-ray photoelectron spectroscopy"** Phys. Rev. Res. 4, 033108 (2022), https://doi.org/10.1103/PhysRevResearch.4.033108 .

Falke J., et al.**"Electronic structure of the metallic oxide ReO_3"** Phys. Rev. B 103, 115125 (2021), https://doi.org/10.1103/PhysRevB.103.115125 .

[Preprints / Under Review]

Chen, C.S., et al. "A Unified SPD Token Transformer Framework for EEG Classification: Systematic Comparison of Geometric Embeddings" arXiv preprint (2026), arxiv:2601.21521.

Chen, C.S., et al. "FreqLens: Interpretable Frequency Attribution for Time Series Forecasting" arXiv preprint (2026), 2602.08768.

Chen, C.S., et al. "Quantum and Classical Machine Learning in Decentralized Finance: Comparative Evidence from Multi-Asset Backtesting of Automated Market Makers" arXiv preprint (2025), arXiv:2510.15903.

Chen, C.S., et al. "Large Cognition Model: Towards Pretrained EEG Foundation Model" arXiv preprint (2025), arXiv:2502.17464.

Chen, C.S., et al. "Mind's Eye: Image Recognition by EEG via Multimodal Similarity-Keeping Contrastive Learning" arXiv preprint (2024), arXiv:2406.16910.

Professional Service

Program Committee: QNRL Workshop @ IEEE WCCI 2026

Conference Reviewer (2026–2027): **NeurIPS**, **ICML** (Gold Reviewer — top tier), **AAAI**, **KDD**, **MICCAI**, **ICASSP**; NLDL 2027; IEEE QAI 2026

IOP Trusted Reviewer (2026), recognized by the Institute of Physics Publishing for quality and commitment in peer review.

Journal Reviewer (2025–present): **Oxford National Science Review (NSR)**, **IEEE Trans. Pattern Analysis and Machine Intelligence (TPAMI)**, **IEEE Trans. Neural Networks and Learning Systems (TNNLS)**, **IEEE Trans. Automation Science and Engineering (TASE)**, **IEEE Trans. Neural Systems & Rehabilitation Engineering (TNSRE)**, **IEEE Trans. Systems, Man and Cybernetics (TSMC)**, IEEE Trans. Human-Machine Systems (THMS), **IEEE Internet of Things Journal (IoT-J)**, IEEE Trans. Consumer Electronics, **IEEE/ACM Trans. Audio, Speech, and Language Processing (TASLP)**, IEEE Trans. Cognitive and Developmental Systems (TCDS), **IEEE Journal of Biomedical and Health Informatics (JBHI)**, BMC Medical Informatics and Decision Making, IEEE Signal Processing Letters (SPL), IEEE Access, **ACM Trans. Recommender Systems (TORS)**, Tsinghua Science and Technology, **npj Quantum Information**, **Scientific Reports (Nature Portfolio)**, EPJ Quantum Technology, IOP Journal of Neural Engineering, Taylor & Francis Behaviour & Information Technology (BIT), Springer International Journal of Machine Learning and Cybernetics, Springer Cognitive Neurodynamics, Springer Journal of Bionic Engineering, IOP Biomedical Physics & Engineering Express, IOP Engineering Research Express, AIMS Quantitative Finance and Economics, AIMS Public Health, IntechOpen AI, Computer Science and Robotics Technology, Springer Discover Artificial Intelligence, Frontiers in Surgery, Journal of Medical Internet Research (JMIR), Elsevier Neural Networks

Awards & Achievements

2026 NVIDIA GTC Top 8 Finalist: "GPU-Accelerated Chinese Brain-to-Text Interface Powered by NVIDIA H100 80GB."

2026 Berkeley RDI AgentX-AgentBeats: Phase 1 Healthcare Agent 3rd Place (Team MadGAA, 1,300+ teams worldwide).

2023 Certificate of the 20th National Innovation Award, clinical research category.

2019 International Blockchain Olympiad (IBCOL): World Finalist competition with 100+ teams.

2017 Microsoft International Imagine Cup : 2nd place in Taiwan.

2016 International Genetically Engineered Machine Competition (iGEM): World Gold medal, Best Applied Design, Best Part Collection, Best Presentation competition with 300+ university teams.